

Nicholas Amadeus

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EDUCATION

University of Michigan

B.S. Mathematics || B.S. Statistics || B.S. Economics

- Quantitative and Computational Finance Group (QCF) – Quant Developer
- MIG Annual Quant Competition – Rank: 1st place champion / ± 120 participants

Technical Skills: Python, R, SQL, C++, MATLAB | Libraries: pandas, NumPy, scikit-learn, QuantLib, PyTorch

Relevant Coursework:

Math: Stochastic Processes, Probability Theory, Mathematical Modelling, Differential Equations, Financial Mathematics, Real Analysis, Linear and Abstract Algebra, Differential Geometry, Data-Guided Controls, Topological Data Analysis

Statistics: Regression Analysis, Machine Learning, Statistical Computing, Theoretical Statistics, Bayesian Inference

Economics: Game Theory, Microeconomic Theory, Macroeconomic Theory, Econometrics

Ann Arbor, MI

Expected Dec 2026

EXPERIENCE

Math Researcher, University of Michigan Computational Medicine

Optimal Transport Theory and Cellular Reprogramming

- Modeled cell identity transformations as probabilistic mass transport between gene-expression distributions
- Used Wasserstein metrics and entropic regularization to characterize efficient transformations between cellular states

Math Researcher, University of Michigan Math Department

Large-Scale Geometries of the Integers

- Developed algorithms to compute word-lengths for the word metric on the 4×4 Heisenberg group over $\mathbb{Z}$
- Developing algorithms toward proving an isomorphism between algebraic groups $\mathrm{RI}(G, d)$ and $\mathbb{Z}/2\mathbb{Z}$

Data Analyst and Internal Audit Intern, Garuda Metalindo

Operations Analytics and Inventory Systems

- Built C++ and Python ETL tooling to streamline inventory data processing across 2.5B+ items and 50k+ variants/year.
- Developed Python-based anomaly/fraud detection to flag suspicious shortages, accounting discrepancies, and production measurement errors.
- Applied SQL and relational modeling to analyze production/finance tables and design more efficient reporting and validation workflows.

Michigan Math Laboratory Staff

Tutor and Proctor

- Provided individualized tutoring in calculus maintained a secure testing environment for 700+ students/week

Deloitte Indonesia

Risk Advisory Intern (High School)

- Reviewed bank-client risk database reports to identify compliance issues, and helped draft HR risk assessment controls to reduce exposure to employee-liability and compensation claims.

Ann Arbor, MI

May 2025 – Present

Ann Arbor, MI

Aug 2024 – Present

Jakarta, Indonesia

Jun 2024 – Jul 2024

Ann Arbor, MI

Sep 2023 – Present

Jakarta, Indonesia

Feb 2018 – Mar 2018

PROJECTS

Bayesian Multi-Armed Bandit Portfolio Builder and Optimizer

December 2025

- Modeled daily portfolio construction as a multi-armed bandit problem; included diversified/factor-tilted arms and a Markowitz-style mean-variance portfolio to balance return vs. risk.
- Implemented Thompson sampling with a risk-aware variant that ranks bandit arms by predicted tail risk (CVaR at 5%) to reduce drawdowns and volatility while preserving gains, achieving a 1.104 Sharpe with -0.203 max drawdown

Prediction of Bitcoin Limit Order Book Dynamics via SVMs and Tree Ensembles

November 2025

- Reconstructed level-2 BTCUSDT limit order book and engineered microstructure features (spread/midprice dynamics, depth/imbalance metrics, momentum, short-horizon volatility) to predict next 1-second midprice direction.
- Trained SVMs, Random Forests, XGBoost, and LightGBM with chronological split + walk-forward validation; achieved up to 0.705 out-of-sample directional accuracy with boosted trees.
- Converted predictions into an long/short rule that outperformed buy-and-hold; best model delivered 26.76% total return with Sharpe 0.53 and max drawdown -3.39% .

Predicting Major Depressive Disorder via Statistical Ensembles

April 2025

- Built a classification pipeline on NHANES survey data to identify risk of Major Depressive Disorder and benchmarked model performance across approaches. Used Naive Bayes, LASSO logistic regression, and trees/forest models.

Word Metric Approximation for the 4×4 Heisenberg Group

December 2024

- Designed and proved correct algorithms to compute word metric on $\mathbb{Z}$ and the 4×4 integer Heisenberg group on Python
- Derived an efficient closed-form procedure using Blachère's $H(3)$ length formula and an analytic optimization step to split the central coordinate, yielding an $O(n)$ -time, $O(1)$ -space routine.